The Hillingdon Hospitals
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Antibiotic Guidelines – Obstetrics

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Change History

Version	Author	Summary of changes
2.0	Dr Brama Hanumunthadu, Consultant Microbiologist Mr Orfeas Keramidas, Consultant in Obstetrics and Gynaecology Ramiz Amin, Antibimicrobial Pharmacist	Full review, updates made throughout in line with guidance from the antimicrobial stewardship group
3.0	Orfeas Keramidas Dilan Chauhan, Lead Midwife for QI, Guidelines & Audit	Consistency changes to intrapartum prophylaxis to align with GBS GL (No.627).
3.1	Orfeas Keramidas Dilan Chauhan, Lead Midwife for QI, Guidelines & Audit	Consistency changes to intrapartum prophylaxis to align with PROM GL (No.629).

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Aim/Purpose

The Guidelines helps the physician to choose the most appropriate, evidence based antibiotic for commencing empiric therapy.

Introduction

In 2019 the UK published its 20-year vision for antimicrobial resistance (AMR). This set the ambitious goal of ensuring AMR will be controlled and contained by 2040. The first 5-year national action plan for antimicrobial resistance, 'Tackling antimicrobial resistance 2019 to 2024', was an important step towards achieving this vision.
This national action plan (NAP), 'Confronting antimicrobial resistance 2024 to 2029', builds on the achievements and lessons of the first. It contains outcomes and commitments that will make progress towards the 20-year vision for AMR to be contained, controlled and mitigated.
Antimicrobial stewardship aims to improve the safety and quality of patient care and to contribute significantly to reductions in the emergence and spread of AMR. These aims are ultimately achieved by improving antimicrobial prescribing through an organised antimicrobial management program. A ***Start Smart - then Focus*** approach is recommended for all antibiotic prescriptions.

Start Smart - this means:

- do not start antimicrobial therapy unless there is clear evidence of infection
- take a thorough drug allergy history
- initiate prompt effective antibiotic treatment within one hour of diagnosis (or as soon as possible) in patients with severe sepsis or life-threatening infections. Avoid inappropriate use of broad-spectrum antibiotics
- comply with local antimicrobial prescribing guidance (all doses stated in these guidelines are based on the **assumption of normal renal / liver functions**. If patient presented with renal or hepatic impairment, please refer to online BNF or the datasheets for dose information). Discuss with team consultant / consultant microbiologist, if the suggested antibiotic deemed not suitable due to allergy, comorbidities, drug interaction, contraindication,.....)
- take a thorough medication history and consider possible drug interactions
- consider the risk of resistant pathogens such as **MRSA, ESBL, VRE**, etc producing organisms and when suspected / confirmed **offer alternative regimens** accordingly or contact consultant microbiologist for advice.
- document clinical indication (and disease severity if appropriate), drug name, dose and route on drug chart and in clinical notes*
- include review/stop date or duration
- obtain cultures prior to commencing therapy where possible (but do not delay therapy)
- document the exact indication in the patients electronic structure notes.

Then Focus – this means:

Reviewing the clinical diagnosis and the continuing need for antibiotics at 48-72 hours and documenting a clear plan of action in the patients electronic structure notes. The **'antimicrobial prescribing decision' options** are:

- **Stop** antibiotics if there is no evidence of infection
- **Switch** antibiotics from intravenous to oral
- **Change** antibiotics – ideally to a narrower spectrum – or broader if required
- **Continue** and document next review date or stop date

It is essential that the review and subsequent decision is clearly documented in the clinical notes and on the drug chart where possible eg stop antibiotic

The Guidance

ANTIBIOTIC GUIDELINES - OBSTETRICS

Surgical Prophylaxis for Caesarean Section

Give a single dose of antibiotics to the mother ***before*** skin incision

(antibiotic prophylaxis is most effective when given before skin incision, however if urgency does not permit use clinical judgement and give as quickly as possible)

Cefuroxime(IV) 1.5g + **Metronidazole(IV)** 500mg

Penicillin allergy:

Give a single dose of antibiotics to the mother ***before*** skin incision

Clindamycin(IV) 900mg + **Gentamicin(IV)** 5mg/kg (max 500mg)
(Use actual body weight at booking, unless obese in which case use dose determining weight, refer to "Adult Antibiotic Therapeutic Dosing & Monitoring Guide" for full guidance on dosing and monitoring)

If MRSA positive or history of MRSA in the last 12 months:

Add single dose **Teicoplanin (IV)** 6mg/kg to the above regimen (round up to nearest 100mg based on booking weight)

Reference:

National Institute for Health and Care Excellence. (2024). Caesarean birth. London: National Institute for Health and Care Excellence. NG192. https://www.nice.org.uk/guidance/ng192

Maternal Protocol for Prevention of Group B Strep (GBS) Disease

(If fever or signs of active infection present or develop follow Pyrexia in labour guidelines)

Intrapartum prophylaxis should be offered to women with:

- Vaginal swab confirming GBS colonisation in the current or any previous pregnancies
- Group B streptococcal colonisation, bacteriuria or infection in the current pregnancy
- Previous baby with neonatal GBS disease
- Preterm labour <37 weeks (If <34+6 weeks -should also receive antenatal steroids)

Benzyl penicillin(IV) 3g stat in labour, then 1.5g 4 hourly *until delivery*

Non-anaphylactoid penicillin allergy (i.e. rash, forgotten childhood history of allergy or not known)

Cefuroxime (IV) 1.5g stat in labour, then 750mg every 8 hours *until delivery*

Anaphylactoid penicillin allergy (i.e. anaphylaxis, angioedema, respiratory distress or urticaria)

Teicoplanin (IV) starting in labour 6mg/kg bd for 3 doses, then od *until delivery*
(round up to nearest 100mg based on booking weight)

Reference:

Royal College of Obstetricians & Gynaecologists: Green-top guideline 36, september 2017; 'Prevention of Early-onset Neonatal Group B Streptococcal Disease. https://www.rcog.org.uk/en/guidelines-research-services/guidelines/gtg36/

Preterm Premature Rupture of Membranes(PPROM) (<37 weeks)

(If fever or signs of active infection present or develop follow Pyrexia in labour guidelines)

(Avoid Co-amoxiclav due to increased risk of necrotising enterocolitis in newborn)

1st line

Erythromycin(PO) 250mg 6 hourly for 10 days or until established labour (whichever is sooner)

(Associated with a statistically significant delay in delivery and reduction in neonatal morbidity)

2nd line (if erythromycin contraindicated)

Amoxicillin(PO) 500mg TDS for 10 days or until established labour (whichever is sooner)

Reference:

National Institute for Health and Care Excellence. (updated 2022). Preterm labour and birth. London: National Institute for Health and Care Excellence. NG25. https://www.nice.org.uk/guidance/ng25

Preterm Labour (<37 weeks):

(If fever or signs of active infection present or develop follow Pyrexia in labour guidelines)

Benzyl penicillin(IV) 3g stat in labour, then 1.5g 4 hourly *until delivery*

Non-anaphylactoid penicillin allergy (i.e. rash, forgotten childhood history of allergy or not known)

Cefuroxime (IV) 1.5g stat in labour, then 750mg every 8 hours *until delivery*

Anaphylactoid penicillin allergy (i.e. anaphylaxis, angioedema, respiratory distress or urticaria)

Teicoplanin (IV) starting in labour 6mg/kg bd for 3 doses, then od *until delivery*
(round up to nearest 100mg based on booking weight)

Reference:

National Institute for Health and Care Excellence. (2024). Neonatal infection: antibiotics for prevention and treatment. London: National Institute for Health and Care Excellence. NG195.
https://www.nice.org.uk/guidance/ng195

Pre labour rupture of membranes at term (PROM)

(If fever or signs of active infection present or develop follow Pyrexia in labour guidelines)

Determine if patient meets criteria for GBS prophylaxis – if criteria met, Intrapartum prophylaxis for prevention of early onset GBS should start as soon as rupture of membranes confirmed

Additionally, where prelabour rupture of membranes >24 hours, for intrapartum prophylaxis only please follow the below schedule:

Benzyl penicillin(IV) 3g stat in labour, then 1.5g 4 hourly *until delivery*

Non-anaphylactoid penicillin allergy (i.e. rash, forgotten childhood history of allergy or not known)

Cefuroxime (IV) 1.5g stat in labour, then 750mg every 8 hours *until delivery*

Anaphylactoid penicillin allergy (i.e. anaphylaxis, angioedema, respiratory distress or urticaria)

Teicoplanin (IV) starting in labour 6mg/kg bd for 3 doses, then od *until delivery*
(round up to nearest 100mg based on booking weight)

Reference:

National Institute for Health and Care Excellence. (2024). Neonatal infection: antibiotics for prevention and treatment. London: National Institute for Health and Care Excellence. NG195.
https://www.nice.org.uk/guidance/ng195

Management of 3rd and 4th degree tears

(Broad spectrum antibiotics is recommended following obstetric anal sphincter repair to reduce incidence of postoperative infections and wound dehiscence)

Co-amoxiclav(IV)1.2g in theatre
followed by
Co-amoxiclav(PO) 625mg tds for 5 -7 days (total)

Penicillin allergy:

Clindamycin(IV) 900mg in theatre **+** **Gentamicin(IV)** 5mg/kg (max 500mg) in theatre (Use actual body weight at booking, unless obese in which case use dose determining weight, refer to "Adult Antibiotic Therapeutic Dosing & Monitoring Guide" for full guidance on dosing and monitoring)

followed by

Clindamycin(PO) 450mg qds for 5 -7 days (total)

If MRSA positive or history of MRSA in the last 12 months:

Add Teicoplanin (IV) 6mg/kg to the above regimen in theatre (round up to nearest 100mg based on booking weight) and discuss with microbiologist on next working day

Reference:

Royal College of Obstetricians & Gynaecologists: Green-top guideline 29, June 2015; The Management of Third- and Fourth-Degree Perineal Tears. https://www.rcog.org.uk/en/guidelines-research-services/guidelines/gtg29/

Manual removal of the placenta

Give a single dose of antibiotics to the mother intraoperatively

Cefuroxime(IV) 1.5g **+** **Metronidazole(IV)** 500mg

Penicillin allergy:

Give a single dose of antibiotics to the mother intraoperatively

Clindamycin(IV) 900mg **+** **Gentamicin(IV)** 5mg/kg (max 500mg)
(Use actual body weight at booking, unless obese in which case use dose determining weight, refer to "Adult Antibiotic Therapeutic Dosing & Monitoring Guide" for full guidance on dosing and monitoring)

If MRSA positive or history of MRSA in the last 12 months:

Add single dose **Teicoplanin (IV)** 6mg/kg to the above regimen (round up to nearest 100mg based on booking weight)

Cervical cerclage

Routine prophylaxis is not recommended

Reference:

Royal College of Obstetricians & Gynaecologists: Green-top guideline 75, June 2022; Cervical cerclage.
https://obgyn.onlinelibrary.wiley.com/doi/epdf/10.1111/1471-0528.17003

Operative vaginal delivery

Co-amoxiclav(IV) 1.2g single dose as soon as possible **after** giving birth.

Penicillin allergy: **Clindamycin(IV)** 900mg single dose as soon as possible **after** giving birth.

If MRSA positive or history of MRSA in the last 12 months:

Add single dose **Teicoplanin (IV)** 6mg/kg to the above regimen (round up to nearest 100mg based on booking weight)

Reference:

Royal College of Obstetricians & Gynaecologists: Green-top guideline 26, April 2020; Assisted Vaginal Birth.
https://obgyn.onlinelibrary.wiley.com/doi/10.1111/1471-0528.16092

Pyrexia in Labour/Suspected Chorioamnionitis (Pyrexia ≥ 38ºC)

Take blood cultures, vaginal swab, urine sample, FBC and CRP.

Cefuroxime(IV) 1.5g tds	+	**Metronidazole(IV)** 500mg tds

Penicillin allergy:

Teicoplanin(IV) 6mg/kg bd for 3 doses, then od rounded up to nearest 100 mg based on booking weight Refer to "Adult Antibiotic Therapeutic Dosing & Monitoring Guide" for full guidance on dosing and monitoring)	+	**Metronidazole(IV)** 500mg tds	+	**Gentamicin(IV)** 5mg/kg od (max 500mg) Use actual body weight at booking, unless obese in which case use dose determining weight, refer to "Adult Antibiotic Therapeutic Dosing & Monitoring Guide" for full guidance on dosing and monitoring)

- Initial Review at 24 then 48-72 hours with micro results
- Confirmed Chorioamnionitis /puerperal sepsis treat for 7 days in total (including oral switch)
- IV to oral switch if apyrexial for 24-48 hours and clinically appropriate

Reference:

Royal College of Obstetricians & Gynaecologists: Green-top guideline 64a, April 2012; Bacterial Sepsis in Pregnancy. https://www.rcog.org.uk/en/guidelines-research-services/guidelines/gtg64a/

Nonviable fetus, late TOP or IUD with pyrexia

Women with sepsis should be treated with intravenous broad-spectrum antibiotic therapy (including antichlamydial agents) and test for PID

Cefuroxime(IV)	**+**	**Metronidazole(IV)**	**+/-**	**Gentamicin(IV)**
1.5g tds		500mg tds		5mg/kg od (max 500mg) Use actual body weight at booking, unless obese in which case use dose determining weight, refer to "Adult Antibiotic Therapeutic Dosing & Monitoring Guide" for full guidance on dosing and monitoring)

Plus

Azithromycin(PO) 1g once a week for 2weeks

If severe sepsis (fever >38ºC, tachycardia, hypotension, lactate >2)

Piperacillin/tazobactam (IV)	**+/-**	**Gentamicin(IV)**
4.5g qds		5mg/kg od (max 500mg) Use actual body weight at booking, unless obese in which case use dose determining weight, refer to "Adult Antibiotic Therapeutic Dosing & Monitoring Guide" for full guidance on dosing and monitoring)

Plus

Azithromycin(PO) 1g once a week for 2weeks

If Penicillin allergy:

Clindamycin(IV)	**+**	**Gentamicin(IV)**
1.2g qds		5mg/kg od (max 500mg) Use actual body weight at booking, unless obese in which case use dose determining weight, refer to "Adult Antibiotic Therapeutic Dosing & Monitoring Guide" for full guidance on dosing and monitoring)

Plus

Azithromycin(PO) 1g once a week for 2weeks

If MRSA positive or history of MRSA in the last 12 months:

Add Teicoplanin (IV) 6mg/kg bd for 3 doses, then od

to the above regimen (round up to nearest 100mg based on booking weight)

Refer to "Adult Antibiotic Therapeutic Dosing & Monitoring Guide" for full guidance on dosing and monitoring)

Reference:

Royal College of Obstetricians & Gynaecologists: Green-top guideline 55, 2024; Care of late intrauterine fetal death and stillbirthhttps://obgyn.onlinelibrary.wiley.com/doi/epdf/10.1111/1471-0528.17844

Pyrexia / Retained Products post delivery (Pyrexia ≥ 38ºC)

(If pelvic sepsis is suspected prompt early treatment with a combination of high-dose broad-spectrum intravenous antibiotics may be lifesaving).

Use a different antibiotic from any antibiotic treatment or prophylaxis has already been given at time of labour

Cefuroxime(IV) 1.5g tds	+	**Metronidazole(IV)** 500mg tds	+/-	**Gentamicin(IV)** 5mg/kg od (max 500mg) Use actual body weight at booking, unless obese in which case use dose determining weight, refer to "Adult Antibiotic Therapeutic Dosing & Monitoring Guide" for full guidance on dosing and monitoring).

If severe sepsis (fever >38ºC, tachycardia, hypotension, lactate >2)

Piperacillin/tazobactam (IV) 4.5g qds	+/-	**Gentamicin(IV)** 5mg/kg od (max 500mg) Use actual body weight at booking, unless obese in which case use dose determining weight, refer to "Adult Antibiotic Therapeutic Dosing & Monitoring Guide" for full guidance on dosing and monitoring).

Penicillin allergy:

Clindamycin(IV) 1.2g qds	+	**Gentamicin(IV)** 5mg/kg od (max 500mg) Use actual body weight at booking, unless obese in which case use dose determining weight, refer to "Adult Antibiotic Therapeutic Dosing & Monitoring Guide" for full guidance on dosing and monitoring).

If MRSA positive or history of MRSA in the last 12 months:

Add Teicoplanin (IV) 6mg/kg bd for 3 doses, then od
to the above regimen (round up to nearest 100mg based on booking weight)
Refer to "Adult Antibiotic Therapeutic Dosing & Monitoring Guide" for full guidance on dosing and monitoring)

- The expert advice of a consultant microbiologist should be sought at an early stage
- Review microbiology results as soon as possible.

If patient apyrexial for 24-48 hours, change to oral antibiotics taking in consideration available microbiology results or seek advice from consultant microbiologists.

Reference:
Royal College of Obstetricians & Gynaecologists: Green-top guideline 64b, April 2012; Bacterial Sepsis following Pregnancy. https://www.rcog.org.uk/en/guidelines-research-services/guidelines/gtg64b/

Urinary Tract Infections in Pregnancy

1. Uncomplicated UTI

- take urine sample prior to starting antibiotics if possible
- treatment should be guided by MC&S whenever possible

1st line (Up to 36 weeks):

Nitrofurantoin(PO) 50mg qds for 7 days
(Nitrofurantoin may produce neonatal haemolysis if used at term - avoid after 36 weeks pregnancy or if delivery suspected within one week)

2nd line:

Cefalexin(PO) 500mg bd for 7 days

3rd line

Co-amoxiclav(PO) 625mg tds for 7 days
(Risk of necrotising enterocolitis in neonate - avoid if PROM or preterm labour <36 weeks)

Note

- Do not use amoxicillin or trimethoprim empirically due to high resistance rates locally. Amoxicillin and trimethoprim are good choices **if sensitivity known**
- Trimethoprim should be avoided in the first trimester due to anti-folate activity. If trimethoprim is used in pregnancy, folic acid 5mg once daily should also be given.

Reference

NICE 2018. NG109: Urinary tract infection (lower): antimicrobial prescribing https://www.nice.org.uk/guidance/ng109

2. Pyelonephritis or systemic involvement

- take urine sample prior to starting antibiotics if possible
- treatment should be guided by MC&S whenever possible.
- continue antibiotics for 14 days)

Cefuroxime(IV) 1.5g tds	**+/-**	**Gentamicin(IV)** 5mg/kg od (max 500mg) • Review gentamicin after 48 hours • Use actual body weight at booking, unless obese in which case use dose determining weight, refer to "Adult Antibiotic Therapeutic Dosing & Monitoring Guide" for full guidance on dosing and monitoring)

or

Co-amoxiclav(IV) 1.2g tds (Risk of necrotising enterocolitis in neonate - avoid if PROM or preterm labour <36 weeks)	**+/-**	**Gentamicin(IV)** 5mg/kg od (max 500mg) • Review gentamicin after 48 hours • Use actual body weight at booking, unless obese in which case use dose determining weight, refer to "Adult Antibiotic Therapeutic Dosing & Monitoring Guide" for full guidance on dosing and monitoring)

Severe penicillin allergy:

Gentamicin(IV) 5mg/kg od (max 500mg)
(Use actual body weight at booking, unless obese in which case use dose determining weight, refer to "Adult Antibiotic Therapeutic Dosing & Monitoring Guide" for full guidance on dosing and monitoring)

If history / risk of ESBL follow sensitivities or discuss with micro

Meropenem (IV) 1g tds and discuss with microbiologist in normal working hours

Reference

NICE 2018. NG111: Pyelonephritis (acute): antimicrobial prescribing https://www.nice.org.uk/guidance/ng111

3. Recurrent UTI in pregnancy

- Women who have been advised to take long-term antibiotic prophylaxis for recurrent UTI pre-pregnancy should continue it in pregnancy.
- Women with two or more confirmed UTI in pregnancy should have an ultrasound scan of the renal tract, and antibiotic prophylaxis until delivery should be considered
- Suitable regimens include:
    - **Nitrofurantoin(PO)** 50-100mg nocte
      (May produce neonatal haemolysis if used at term - avoid after 37 weeks pregnancy or if delivery suspected within one week)
    - **Cefalexin(PO)** 125 - 250mg nocte

- Reassess patient after 8 weeks of antibiotic treatment and consider stopping or change of antibiotic.

Reference

NICE 2018. NG112: Urinary tract infection (recurrent): antimicrobial prescribing
https://www.nice.org.uk/guidance/ng112

Associated Guidelines, Polices and Documents

Antibiotic policy (overarching)
Prescribing, Administration and Security of Medicines policy
Penicillin Allergy Poster
Antimicrobials Categories List

Training Need Identified (if yes, details)

Staff will be made aware of guidelines during the induction process by their clinical line manager. Staff has a responsibility to check the intranet for the latest version of guidelines when and as required by clinical practice.

Updates to guidelines are notified according to the process described in section 6. There is no specific training course associated with this guideline.

Implementation and Monitoring

All Guidelines are available on the intranet. All staff has a responsibility to check the Trust's intranet and ensure that they are referring to the most recent issue of the document.

New and updated versions of the guideline are implemented as follows:

- Notifications automatically generated and pushed to mobile devices whenever the guidelines on the mobile devices application are updated.
- Dissemination of the guidelines to divisions through the divisional representation to the Antimicrobial stewardship Group
- The weekly general information bulletin includes notification of publication of new/updated documents.

This clinical guideline will be monitored using local guidance and via the Trust Incident and Risk Management Policies. If risks/issues are identified further specific monitoring will be considered, for example, clinical audit or risk assessment.

References

Please see under each section.

Appendices

None.